

Faculty of Computing

Year 2 Semester 1 (2026)

IT2011 - Artificial Intelligence and Machine Learning

Final Evaluation - Implementation

This stage focuses on Model Design, Implementation, and Evaluation Preparation using your assigned dataset, and will be assessed through a viva presentation.

Progress Review II – Viva Details

- Mode: In-class (during scheduled lab/workshop sessions)
- Duration: 15 minutes per student
- Format: Viva-style explanation and demonstration of models

Individual Requirements (55 marks per student)

Each student must:

- Design and implement at least one machine learning model, such as:
 - Classification (e.g., SVM, Decision Tree, Logistic Regression)
 - Clustering (e.g., K-Means, DBSCAN)
 - Dimensionality Reduction (e.g., PCA)
 - Deep Learning (e.g., MLP, CNN using Keras/TensorFlow)

Explain your model by covering:

1. Model suitability to your dataset and problem
2. Implementation details (libraries, functions, hyperparameters)
3. Parameter tuning methods (e.g., GridSearchCV, manual tuning)
4. The number and variety of the models trained by tuning parameters
5. The evaluation metric(s) for each variety of the model trained:
 - e.g., Accuracy, F1 Score, Precision, Recall, Confusion Matrix, AUC, etc.
 - Why these metrics are appropriate for the dataset/problem type
 - Validation method used (e.g., train-test split, k-fold cross-validation)
 - A comparison between two or more varieties of the models with a conclusion
 - Any limitations, observations, or improvements suggested